

NATHAN DURAND

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EDUCATION

Iowa State University

Bachelor of Science in Mechanical Engineering | GPA: 3.53 / 4.0

Ames, Iowa

August 2024 – May 2028

- **Honors:** Dean's List, NSBE Honeywell IPP Scholar, UNCF ETOP Scholar

WORK EXPERIENCE

Honeywell Aerospace

Aerospace Engineering Intern

Phoenix, AZ

May 2026 – August 2026

- Conducted R&D testing and failure analysis on engines and other aerospace hardware in a high-altitude cold chamber simulating flight conditions; performed root-cause investigations to enhance design reliability and safety compliance.
- Applied Lean Six Sigma methodologies to F-35, F-16, and F-15 component production lines through cross-functional collaboration, identifying bottlenecks and improving manufacturing throughput.
- Optimized supply chain performance for Airbus A320 and Boeing 737 platforms via proactive capacity analysis and direct process-improvement initiatives with external suppliers.
- Developed engineering proposals for space-based data center architectures; evaluated integration of Control Moment Gyroscopes (CMG) for spacecraft attitude control to maximize solar power capture and optical laser communication alignment.

Turner Construction (Meta)

Engineering Intern

Omaha, NE

May 2025 – August 2025

- Coordinated daily operations for 200+ trade workers on a \$500M+ data center construction project, ensuring schedule compliance and resolving on-site engineering challenges in real time.
- Reviewed engineering drawings, led multidisciplinary coordination meetings, and authored weekly progress and safety reports for project management and ownership teams.
- Worked in the mechanical yard overseeing installation and coordination of HVAC and water cooling systems, supporting the mechanical infrastructure for a hyperscale data center facility.

TD2 Engineering and Surveying

Engineering Intern

Omaha, NE

September 2023 – August 2024

- Earned FAA Part 107 Unmanned Aircraft Systems (UAS) pilot certification; led UAV-based aerial site surveys that improved land data accuracy and efficiency for civil development projects.
- Performed ASTM-compliant materials testing on concrete and soil specimens, analyzing compressive strength, composition, and structural integrity to support quality control and project reliability.
- Managed independent project assignments and mentored incoming interns on field procedures, data collection, and engineering documentation standards.

ENGINEERING PROJECTS

Hand Crank Grain Mill — Off-Grid Humanitarian Design | SolidWorks, QFD, DFMEA

Jan. 2026 – May 2026

- Designed a manually powered grain mill for off-grid rural communities in Nigeria, targeting under \$50 and 15 lbs.
- Validated design via QFD, DFMEA, and three models (crank torque: 66.61 lbf-in; 376.99 rad/min; 3 lb/min milling rate).
- Fabricated an 8-subassembly plywood prototype from initial sketch using a table saw, miter saw, and drill press; presented at the Iowa State University Design Expo.

Assistive Bowling Device for Children with Special Needs | Autodesk Inventor, 3D Printing

Sep. 2023 – May 2024

- Designed and 3D-printed a switch-activated mechanical device enabling individuals with motor impairments to bowl independently; donated to a local special-needs school serving wheelchair-dependent children.
- Gathered iterative user feedback and implemented design improvements to enhance usability, accessibility, and performance through human-centered engineering methodology.

TECHNICAL SKILLS & CERTIFICATIONS

Software / CAD: SolidWorks, Fusion 360, Autodesk Inventor, MATLAB, Python, Revit, Excel

Engineering Skills: Failure Analysis, Stress Testing, Materials Testing, Additive Manufacturing (3D Printing), Lean Six Sigma, UAV/Drone Operations, Root-Cause Analysis, GEO Mapping

Certifications: FAA Part 107 Unmanned Aircraft Systems (UAS) Pilot License · OSHA 10

LEADERSHIP & INVOLVEMENT

National Society of Black Engineers (NSBE) — Academic Excellence Chair **SolidWorks Club** — Member

HJJ Athletic Excellence Scholarship Board — Board Member **Campus Tour Guide** — Iowa State University